

Example question sheet

Simple SELECT, WHERE, IS NULL, LIKE, ORDER BY Question 1

1. List the red and white cars that were bought before 2015.
2. List car license plate numbers, where the color is unknown.
3. List the full name of car owners, with their age rounded to 2 decimals and the city they live in. Order the list by their tax number. The city can be extracted from the address.
4. List the full name of car models (Brand and model name, like Volkswagen Golf)
5. List the length of each repair in the database in days. If the work is not finished, the end date is null; use the current date.
6. List cars that were sold for 6 to 7 million forints at their first sale.
7. List car brands and models, where the car brand contains the letter 'o' in lower or uppercase but no 'y', lower or uppercase.
8. List car brands where the name is shorter than or equal to 5 characters.

SELECTs with dates Question 2

1. List cars that were bought in leap years. A leap year is a year in which February is one day longer; for simplicity, take every year that is divisible by 4 (we do not consider the other rules of leap years). 2020 is a leap year, as is 2016, but 2026 is not.
2. Select the car repairs that were started in the morning, 00:00:00-11:59:59.
3. List the mechanics with Leo's zodiac sign. Leos are born between 23rd of July and 22nd of August.
4. List the price of the car evaluations from the last 3 years. Make sure that the query shows the last 3 years whenever they are run.

GROUP BY, HAVING Question 3

1. List the car brand(s) that created the most models. If there are more of them, list all of them.
2. List car colors in the database that are used by fewer than 3 cars.
3. Only identifying them with their ID, how many fulfilled repairs did each workshop finish in the past 5 years? List only the workshops that had more than 20! Order by the number of repairs. The repair is fulfilled if the end date is known.

JOIN with 2 tables Question 4

1. List every car model that has a car.
2. List car models that have no car.
3. List all the car models that were sold in the basic car colors, white, black, and grey, in the last 5 years firstly.
4. List owners who bought cars of unknown colors. Only list the names of the owners, and every name should appear once.
5. List the license plate number, price, brand, model name, and model description of the cars that do not use the standard, 6-character-long license plate.

JOIN with 2 tables, GROUP BY, HAVING Question 5

1. Yellow is considered an unlucky color for cars. List how much money was spent on each yellow car in total in terms of repair in the database.
2. List car owners who have bought more than 3 cars in their lifetime.
3. What is the priciest car of the German car brands? (Volkswagen, Opel, Mercedes-Benz, Audi, BMW) We understand the highest cost in terms of the first selling price.
4. List all the models which make is Volkswagen whose average price is higher than 3 million.
5. What was the highest evaluation of each car in the database? List the car's license plate and its highest evaluation.
6. What was the largest difference between the first sell price and the evaluation price for each car in the database?
7. What is the Volkswagen peak season? List each month and how many Volkswagen s are sold for the first time in those months. List each month ordered by the number of sales descending.

JOIN with 3 tables; OUTER JOIN Question 6

1. Which car model had the longest repair time? List the brand, model, license plate number. and time in days of the car, whose repair time (from start to finish) was the longest.
2. What cars are owned by Papp Mohamed? List their ID, license plate number, color, and price.
3. What car models have never been evaluated? List their make and model.

4. Which mechanics worked for the workshop which name is “Harmat Kft.” in the past 5 years?
5. Which is the latest car owned by Papp Mohamed?

Nested SELECTs with group functions and IN; EXISTS, ALL, ANY (subquery) Question 7

1. List cars that were repaired for a longer time period than the longest repair time of any yellow car. In case a car repair is not finished, use the current date as an end date.
2. Look up cars that are more expensive at their first selling price than any cars with a make of Volkswagen.
3. Which cars were owned by the same owners who had any cars with a make of Opel Astra?
4. Which cars had more repairs than any cars with a make of Volkswagen?
5. Which cars were repaired in the workshop which name is “Harmat Kft.”? List their ID and license_plate_number . Every car should appear once, even if it was repaired multiple times.
6. Which owners got the same type of car as the car that needed to be repaired the most?
7. Which car(s) took the shortest time to repair in the workshop which name is “Harmat Kft”, and what was the cost? Print the car’s details.
8. Which car was the most expensive to repair? Print the license plate number, model, make, and color as well.

NOT IN, UNION, INTERSECT, MINUS; top N Question 8

1. Volkswagen bought Skoda in 2000 and Audi in 2007. How many cars in the database belong to Volkswagen, including Skoda and Audi models after Volkswagen bought them? Count actual cars, not models.
2. List cars bought (based on the first_sell_date) after 2015 that were never owned by people living in Tata (Hungarian town, this is part of their address) and are not diesel (car model’s description does not contain diesel).
3. List diesel cars that are painted red and weren’t repaired more than 10 times, also the gasoline cars that were sold after 2008. In the output list, include the car make and model, license plate number, and color. (diesel car: car model’s description contains diesel, gasoline car: car model’s description contains gasoline).

4. List American cars (which make is Chevrolet or Ford) that were painted white and were bought for under 15 million (by their first-sell-price), and the German cars (which make is Volkswagen, Audi, BMW, Opel, or Mercedes-Benz) that were evaluated under 8 million at least once.

5. List cars, where the highest price of evaluation was more than the total price of their repair. Exclude French cars (their makes are Peugeot, Renault, Dacia).

CREATE TABLE Question 9

1. Create a table named `m_modifications` storing the non-repair type modifications on each car, including the following columns with arbitrary names: `identifier`(number, this is the primary key), `placement`(string, up to 50 characters), `modification name` (string, up to 125 characters), `validation date` (date), `car_id` (number(5)) with reference to the `m_car` table.

2. Create a table named `m_vehicle_inspection` that stores vehicle inspection records for each car. This should include the following columns with arbitrary names: `car_id` (number(5)), `inspection date` (date), `comments`, with the default value of “None” (string, can be up to 255 characters long), and `valid through` (date), which must not be unknown. Name the constraints. The primary key is the combination of `vehicle_id` and `inspection date`.

ALTER TABLE, DROP TABLE; GRANT, REVOKE Question 10

1. Rename the `m_mechanic` table's `birth_date` column to `date_of_birth`.

2. Add a new column named **horsepower** to the `m_car_model` table that stores the horsepower of each model. The value should be a number with up to 4 digits and no decimal places.

3. Drop the primary key of the `m_car_make` table.

4. Add a new column to the `m_owner` table that contains their birth date (date).

5. Grant SELECT privilege to panovics on your `m_car` table.

6. Revoke SELECT privilege from panovics on your `m_car` table.

7. Grant all privileges on your `m_car_mechanic` table to everyone.

CREATE VIEW (easier) Question 12

1. Create a view that lists all the cars which make is Volkswagen with all the columns of both the model and car table.

2. Create a view that calculates the number of owners who have ever owned a car for each car brand.
3. Create a view that displays how many cars were bought by each owner (id and name) after 2015.
4. Create a view that lists all red cars that were evaluated after 2015. Display only the cars' details, not the evaluations. Each car should be displayed only once.

Question 11 contains queries related to DML: INSERT, UPDATE, DELETE